Correlations of the von Mangoldt and higher divisor functions I. Long shift ranges

Terence Tao · 6 July 2017

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Kaisa Matomaki, Maksym Radziwill, and I have uploaded to the arXiv our paper "Correlations of the von Mangoldt and higher divisor functions I. Long shift ranges", submitted to Proceedings of the London Mathematical Society. This paper is concerned with the estimation of correlations such as

$$\sum_{n \leq X} \Lambda(n)\Lambda(n+h) \tag{1}$$

for medium-sized h and large X, where Λ is the von Mangoldt function; we also consider variants of this sum in which one of the von Mangoldt functions is replaced with a (higher order) divisor function, but for sake of discussion let us focus just on the sum (1). Understanding this sum is very closely related to the problem of finding pairs of primes that differ by h; for instance, if one could establish a lower bound

$$\sum_{n \leq X} \Lambda(n)\Lambda(n+2) \gg X$$

then this would easily imply the twin prime conjecture.

The (first) Hardy-Littlewood conjecture asserts an asymptotic

$$\sum_{n \leq X} \Lambda(n)\Lambda(n+h) = \mathfrak{S}(h)X + o(X) \tag{2}$$

as $X \to \infty$ for any fixed positive h, where the *singular series* $\mathfrak{S}(h)$ is an arithmetic factor arising from the irregularity of distribution of Λ at small moduli, defined explicitly by

$$\mathfrak{S}(h) := 2\Pi_2 \prod_{p \mid h; p>2} \frac{p-2}{p-1}$$

when h is even, and $\mathfrak{S}(h) = 0$ when h is odd, where

$$\Pi_2 := \prod_{p>2} (1 - \frac{1}{(p-1)^2}) = 0.66016\ldots$$

is (half of) the twin prime constant. See for instance this previous blog post for a a heuristic explanation of this conjecture. From the previous discussion we see that (2) for $h=2$ would imply the twin prime conjecture. Sieve theoretic methods are only able to provide an upper bound of the form $\sum_{n \leq X} \Lambda(n)\Lambda(n+h) \ll \mathfrak{S}(h)X$.

Needless to say, apart from the trivial case of odd h, there are no values of h for which the Hardy-Littlewood conjecture is known. However there are some results that say that this conjecture holds "on the average": in particular, if H is a quantity depending on X that is somewhat large, there are results that show that (2) holds for most (i.e. for $1 - o(1)$) of the h betwen 0 and H. Ideally one would like to get H as small as possible, in particular one can view the full Hardy-Littlewood conjecture as the endpoint case when H is bounded.

The first results in this direction were by van der Corput and by Lavrik, who established such a result with $H = X$ (with a subsequent refinement by Balog); Wolke lowered H to $X^{5/8+\varepsilon}$, and Mikawa lowered H further to $X^{1/3+\varepsilon}$. The main result of this paper is a further lowering of H to $X^{8/33+\varepsilon}$. In

fact (as in the preceding works) we get a better error term than $o(X)$, namely an error of the shape $O_A(X \log^{-A} X)$ for any A.

Our arguments initially proceed along standard lines. One can use the Hardy-Littlewood circle method to express the correlation in (2) as an integral involving exponential sums $S(\alpha) := \sum_{n \leq X} \Lambda(n) e(\alpha n)$. The contribution of "major arc" α is known by a standard computation to recover the main term $\mathfrak{S}(h)X$ plus acceptable errors, so it is a matter of controlling the "minor arcs". After averaging in h and using the Plancherel identity, one is basically faced with establishing a bound of the form

$$\int_{\beta-1/H}^{\beta+1/H} |S(\alpha)|^2 \, d\alpha \ll_A X \log^{-A} X$$

for any "minor arc" β. If β is somewhat close to a low height rational a/q (specifically, if it is within $X^{-1/6-\varepsilon}$ of such a rational with $q = O(\log^{O(1)} X)$), then this type of estimate is roughly of comparable strength (by another application of Plancherel) to the best available prime number theorem in short intervals on the average, namely that the prime number theorem holds for most intervals of the form $[x, x + x^{1/6+\varepsilon}]$, and we can handle this case using standard mean value theorems for Dirichlet series. So we can restrict attention to the "strongly minor arc" case where β is far from such rationals.

The next step (following some ideas we found in a paper of Zhan) is to rewrite this estimate not in terms of the exponential sums $S(\alpha) := \sum_{n \leq X} \Lambda(n) e(\alpha n)$, but rather in terms of the Dirichlet polynomial $F(s) := \sum_{n \sim X} \frac{\Lambda(n)}{n^s}$. After a certain amount of computation (including some oscillatory integral estimates arising from stationary phase), one is eventually reduced to the task of establishing an estimate of the form

$$\int_{t \sim \lambda X} (\sum_{t-\lambda H}^{t+\lambda H} |\, F(\frac{1}{2} + it')| \, dt')^2 \, dt \ll_A \lambda^2 H^2 X \log^{-A} X$$

for any $X^{-1/6-\varepsilon} \ll \lambda \ll \log^{-B} X$ (with B sufficiently large depending on A).

The next step, which is again standard, is the use of the Heath-Brown identity (as discussed for instance in this previous blog post) to split up Λ into a number of components that have a Dirichlet convolution structure. Because the exponent $8/33$ we are shooting for is less than $1/4$, we end up with five types of components that arise, which we call "Type d_1", "Type d_2", "Type d_3", "Type d_4", and "Type II". The "Type II" sums are Dirichlet convolutions involving a factor supported on a range $[X^\varepsilon, X^{-\varepsilon} H]$ and is quite easy to deal with; the "Type d_j" terms are Dirichlet convolutions that resemble (non-degenerate portions of) the jth divisor function, formed from convolving together j portions of 1. The "Type d_1" and "Type d_2" terms can be estimated satisfactorily by standard moment estimates for Dirichlet polynomials; this already recovers the result of Mikawa (and our argument is in fact slightly more elementary in that no Kloosterman sum estimates are required). It is the treatment of the "Type d_3" and "Type d_4" sums that require some new analysis, with the Type d_3 terms turning to be the most delicate. After using an existing moment estimate of Jutila for Dirichlet L-functions, matters reduce to obtaining a family of estimates, a typical one of which (relating to the more difficult Type d_3 sums) is of the form

$$\int_{t-H}^{t+H} \left| M(\frac{1}{2} + it') \right|^2 \, dt' \ll X^{\varepsilon^2} H \tag{3}$$

for "typical" ordinates t of size X, where M is the Dirichlet polynomial $M(s) := \sum_{n \sim X^{1/3}} \frac{1}{n^s}$ (a fragment of the Riemann zeta function). The precise definition of "typical" is a little technical (because of the complicated nature of Jutila's estimate) and will not be detailed here. Such a claim would follow easily from the Lindelof hypothesis (which would imply that $M(1/2 + it) \ll X^{o(1)}$) but of course we would like to have an unconditional result.

At this point, having exhausted all the Dirichlet polynomial estimates that are usefully available, we return to "physical space". Using some further Fourier-analytic and oscillatory integral computations, we can estimate the left-hand side of (3) by an expression that is roughly of the shape

$$\frac{H}{X^{1/3}} \sum_{\ell \sim X^{1/3}/H} |\sum_{m \sim X^{1/3}} e(\frac{t}{2\pi} \log \frac{m+\ell}{m-\ell})| .$$

The phase $\frac{t}{2\pi} \log \frac{m+\ell}{m-\ell}$ can be Taylor expanded as the sum of $\frac{t_j \ell}{\pi m}$ and a lower order term $\frac{t_j \ell^3}{3\pi m^3}$, plus negligible errors. If we could discard the lower order term then we would get quite a good bound using the exponential sum estimates of Robert and Sargos, which control averages of exponential sums with purely monomial phases, with the averaging allowing us to exploit the hypothesis that t is "typical". Figuring out how to get rid of this lower order term caused some inefficiency in our arguments; the best we could do (after much experimentation) was to use Fourier analysis to shorten the sums, estimate a one-parameter average exponential sum with a binomial phase by a two-parameter average with a monomial phase, and then use the van der Corput B process followed by the estimates of Robert and Sargos. This rather complicated procedure works up to $H = X^{8/33+\varepsilon}$ it may be possible that some alternate way to proceed here could improve the exponent somewhat.

In a sequel to this paper, we will use a somewhat different method to reduce H to a much smaller value of $\log^{O(1)} X$, but only if we replace the correlations $\sum_{n \leq X} \Lambda(n)\Lambda(n+h)$ by either $\sum_{n \leq X} \Lambda(n) d_k(n+h)$ or $\sum_{n \leq X} d_k(n) d_l(n+h)$, and also we now only save a $o(1)$ in the error term rather than $O_A(\log^{-A} X)$.

Comments (9)

Anonymous · 6 July 2017

Dear Terry,
I think there's a typo concerning the value of the twin prime constant, which is around 0.66016.
[Corrected, thanks -T.]

Anonymous · 6 July 2017

Is it possible (by these methods) to improve the new exponent $8/33$? (i.e. is it best possible under current methods ?)

> **Terence Tao** · 6 July 2017
>
> I think there is room for further improvement, in particular new bounds on exponential sums will certainly help (e.g. finding a version of the Robert-Sargos estimates that work in the non-monomial cases). We tried a half a dozen different things while we were at MSRI together and got a whole range of exponents, of which 8/33 was the best that we could come up with, but perhaps there is some arrangement of the existing tools (or, preferably, the introduction of a new tool) that we overlooked. Conjecturally, of course, the exponent should be able to go all the way down to 0. (This is for instance the case if one assumes the exponent pairs conjecture, or even just the Lindelof hypothesis.)

David Cole · 7 July 2017

Good luck with your work on proving the twin prime conjecture! The conjecture is true! Why? Refer to the links below for details.
Note: Proving twin prime, or the more general Polignac conjecture, or Goldbach conjecture depends on the distribution of prime numbers along the natural number line. And if one can prove the Riemann Hypothesis, then will have enough knowledge of the prime distribution to prove those conjectures probabilistically. Prove Riemann Hypothesis and the other results will follow logically.
Reference link:
https://www.quora.com/What-great-conjectures-in-mathematics-combine-additive-theory-of-numbers-with-the-multiplicative-theory-of-numbers/answer/David-Cole-146;
https://www.researchgate.net/post/Why_is_the_Riemann_Hypothesis_true2.

hxypqr · 24 September 2017

it is well known that in the article "A quadratic divisor problem"write by W.Duke,J.B.Friedlander and H.Iwaniec,they proof a linear estimate of this type:$$D_f(a,b;h)=\sum_{am+bn=h}\tau(m)\tau(n)f(am,bn)=\int_0^\infty g(x,x+h)dx+O(P^{\frac{5}{4}}(X+Y)^{\frac{1}{4}}(XY)^{\frac{1}{4}+\epsilon})$$.
based on A.Weil's estimate on Ramanujan sum:$S(m,n;q)=\sum_{d(mod\ q)}e(\frac{dm+\bar{d}n}{q})\leq (m,n,q)^{\frac{1}{2}}q^{\frac{1}{2}}\tau(q)$,and$\delta$ method.and the asymptotic formula $\sum_{n\leq x}\tau(n)\tau(n+h)=xP_h(logx)+O(x^{\frac{11}{12}}log^3x)$.
is it possible to refinement the similar estimate for $H_f(a,b;h)=\sum_{am+bn=h}\Lambda(m)\Lambda(n)f(am,bn)$ use the idea ofδmethod and your so powerful estimate from medium term h in this blog?

> **anonymous** · 24 September 2017
>
> Dear hxypqr,
> The short answer is no, the reason why the method of DFI works for $\tau(n)$ is because $\tau(n)$ has two smooth variables of size at worst $\sqrt{x}$.



hxypqr · 27 September 2017

thank you.yeah you are right,from my calculate i understand the key point is that $\sum_{n\leq x}\Lambda(n)$ does not have a good control with the error term,so because we are deal with the bilinear form,the error term is out of control.in fact if we expect to make the argument still effective.we just need a very good error term estimate of $\sum_{n\leq x}\Lambda(n)$.i think it is nearly to proof RH.

but for the main part,the diagonal estimate and off-diagonal estimate is still effective.



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